

Project 26: Mantello Kia Roxburgh Park – Online Car Sales System

Project Brief:

Business Problem Specifications:

The Mantello's Car Showroom needs to be developed to override the problems prevailing in the practicing manual system. This software is supported to eliminate and in some cases reduce the hardships faced by this existing system. Moreover this system is designed for the particular need of the company to carry out operations in a smooth and effective manner.

The application should reduce as much as possible to avoid errors while entering the data. It should also provide error message while entering invalid data. No formal knowledge should be needed for the user to use this system. Thus by this all it proves it should be user-friendly. Mantello Car Showroom Management System, as described above, should lead to error free, secure, reliable and fast management system. It should assist the user to concentrate on their other activities rather to concentrate on the record keeping. Thus it should help organization in better utilization of resources.

Every organization, whether big or small, has challenges to overcome and managing the information of Customer, Car, Insurance, Booking, Loan. Every Mantello Car Showroom Management System has different Car needs, therefore there is a need to design exclusive employee management systems that are adapted to the managerial requirements. This should be designed to assist in strategic planning, and should help ensure that that the organization is equipped with the right level of information and details for future goals. Also, for those busy executive who are always on the go, the proposed system

should come with remote access features, which will allow the client to manage the workforce anytime, at all times. These systems will ultimately allow the users to better manage resources.

Objective of Project

The main objective of the project is to manage the details of Car, Customer, Payment, Insurance, and Loan. It should manage all the information about Car, Booking, and Loan. The project should totally built at administrative and user ends and only the administrator is guaranteed the complete access. The purpose of the project is to build an application program to reduce the manual work for managing the Car, Customer, Booking, and Payment. It should track all the details about the Payment, Insurance, and Loan.

System Modules

Developers need to research and discuss with the client to finalise the modules and requirements.

User Management Module

Car Inventory and Catalogue Management Module

Search, Filter and Comparison Module

360° Viewer and 3D Car Model Module

Booking and Enquiry Management Module

Payment and Reservation Module

Dealer Management Module Admin

Management Module

Reports and Analytics Module

Notification and Communication Module

QR Code Integration Module

AI Chatbot Module

Input Data and Validation Requirements

- All the fields such as Car, Payment, Loan should be validated and should not take invalid values
- Each form for Car, Customer, Booking should not accept blank value fields
- Avoiding errors in data
- Controlling amount of input
- Integration of all the modules/forms in the system.
- Preparation of the test cases.
- Preparation of the possible test data with all the validation checks.
- Black-box/White-box testing.
- Recording of all the reproduced errors.
- Modifications should be done for the errors found during testing.
- Prepare the test result scripts after rectification of the errors
- Functionality of the entire modules/forms.
- Validations for user input.
- Checking of the Coding standards to be maintained during coding.

- Testing the modules with all the possible test data.
- Testing of the functionality involving all type of calculations.
- Commenting standard in the source files.
- Easy and fast retrieval of information.
- Well-designed reports.
- Decrease the load of the person involve in existing manual system.
- Access of any information individually.
- Work become very speedy.
- Easy to update information

The proposed system should have the following requirements:

- System need stored information about new entry of Car.
- System needs to help the internal staff to keep information of Customer and find them as per various queries
- System need to maintain quantity record.
- System need to keep the record of Payment.
- System need to update and delete the record.
- System also needs a search area.
- It also needs a security system to prevent data.

UI Design

User Interface Design is concerned with the dialogue between a user and the computer. It is concerned with everything from starting the system or logging into the system to the eventually presentation of desired inputs and outputs. The overall flow of screens and messages is called a dialogue.

UI Design Requirements

1. The system user should always be aware of what to do next.
2. The screen should be formatted so that various types of information, instructions and messages always appear in the same general display area.
3. Message, instructions or information should be displayed long enough to allow the system user to read them.
4. Use display attributes sparingly.
5. Default values for fields and answers to be entered by the user should be specified.
6. A user should not be allowed to proceed without correcting an error.
7. The system user should never get an operating system message or fatal error.

The aim of proposed system is to develop a system of improved facilities. The proposed system can overcome all the limitations of the existing system. The system provides proper security and reduces the manual work.

- Security of data.
- Ensure data accuracy's
- Proper control of the higher officials.
- Minimize manual data entry.
- Minimum time needed for the various processing.
- Greater efficiency.
- Better service.
- User friendliness and interactive.
- Minimum time required.

Functional requirements:

Functional requirements are product features or functions that developers must implement to enable users to accomplish their tasks. So, it's important to make them clear for the stakeholders. Generally, functional requirements describe system behavior under specific conditions. The developers of this system must enhance the performance and efficiency of the system by adding 15 to 20 more functional requirements. Students need to do their own research to find how they can improve the system and which FRs need to be added. The group must need a prior approval from the stakeholders/project supervisor before finalizing these Functional Requirements. These enhanced FRs must be reflected separately in Final SRS Report after the approval.

User Management

FR1: The system should allow new users to register with name, email, contact number and password.

FR2: The system should verify users via email before account activation.

FR3: The system should allow users to log in and log out securely. **FR4:** The system should allow users to update their personal profile and password.

FR5: The system should support following three user roles: Customer, Dealer and Administrator.

FR6: The system should display a personalised dashboard for each user role.

FR7: The system should allow customers to view and manage their bookings, enquiries and reservations.

Car Inventory and Catalogue Management

FR8: The system should allow dealers to add new cars with detailed specifications (brand, model, year, price, mileage, transmission, body type, fuel type etc.).

FR9: The system should support uploading multiple high resolution images for each car listing.

FR10: The system should allow dealers to mark cars as *new*, *used*, *reserved* or *sold*.

FR11: The system should validate all required fields before allowing a car listing to be published.

FR12: The system should allow administrators to approve or reject new car listings.

FR13: The system should allow dealers to edit or remove car listings before they are reserved or sold.

Search, Filter and Comparison

FR14: The system should allow users to search for cars by brand, model, year, price, location or body type.

FR15: The system should allow users to apply multiple filters simultaneously.

FR16: The system should display search results in list or grid view with sorting options (price, year, mileage).

FR17: The system should allow users to compare up to three cars side-by-side.

FR18: The system should allow users to add cars to a favourites or watchlist.

FR19: The system should display real-time availability and updated prices for each car.

FR20: The system should allow users to view each vehicle using a 360° image rotation viewer.

FR21: The system should allow selected vehicles to be displayed using an interactive 3D car model.

Booking and Enquiry Management

FR22: The system should allow customers to book a test drive by selecting a date, time and preferred dealer location.

FR23: The system should send booking confirmation and reminder notifications to customers.

FR24: The system should allow users to send enquiries or messages to the dealer regarding specific cars.

FR25: The system should store all enquiries in the database.

FR26: The system should allow dealers to respond to customer enquiries through the system.

FR27: The system should allow admin to view and manage all bookings and enquiries.

Payment and Reservation Module

FR28: The system should allow customers to reserve a car by paying a deposit online.

FR29: The system should integrate secure payment gateways (e.g. PayPal, Stripe, credit card).

FR30: The system should automatically generate invoices for successful reservations.

FR31: The system should store all payment transactions with timestamps. **FR32:** The system should notify the dealer and customer upon successful reservation or payment.

Dealer Management Module

FR33: The system should allow dealers to register and submit verification documents.

FR34: The system should allow admin to approve or reject dealer registration.

FR35: The system should allow dealers to manage their inventory, promotions and pricing.

FR36: The system should provide dealers with performance reports (sales, enquiries, test drives).

Admin Management Module

FR37: The system should allow admin to manage all users, dealers, car listing and reservations.

FR38: The system should allow admin to approve, reject, or remove car listings.

FR39: The system should allow admin to view all transactions and system activity logs.

FR40: The system should allow admin to generate and export system usage reports.

FR41: The system should provide an admin dashboard with KPIs (inventory count, total sales, enquiries, test drives).

Reports and Analytics

FR42: The system should generate reports on top selling car models and brands.

FR43: The system should generate statistics for monthly revenue, test drives and dealer performance.

FR44: The system should allow admin and dealers to export reports as PDF or Excel files.

Notification and Communication Module

FR45: The system should send automated notifications for bookings, payments, responses and status updates.

FR46: The system should display real-time dashboard notifications for all user roles.

FR47: The system should provide live chat capability allowing customers to communicate with car dealers in real time.

QR Code Integration Module

FR48: The system should generate a unique QR code for every vehicle in the inventory.

FR49: The system should allow customers in the physical showroom to scan the QR code to instantly access full car details on their mobile device.

AI Chatbot Module

FR50: The system should include an AI chatbot for assisting customers with car selection.

FR51: The AI chatbot should answer common questions regarding price range, features, brand preferences and car categories.

FR52: The AI chatbot should recommend suitable cars based on user responses and available inventory.

Non-Functional Requirements

Performance Requirements

NFR1 Response Time:

The system should load all main pages (home, listings, car details) within 3 seconds under normal network conditions.

NFR2 Concurrent Users:

The system should support at least 100 simultaneous users browsing or booking without performance issues.

NFR3 Media Performance:

The 360° car viewer and 3D model viewer should load within 5 seconds and operate smoothly at least 25 FPS on standard devices.

Reliability Requirements

NFR4 System Uptime:

The system should maintain a minimum 99% uptime during operational hours.

NFR5 Data Accuracy:

All booking, reservation and transaction records should remain accurate and consistent even during network or server interruptions.

Security Requirements**NFR6 Password Security:**

All user passwords should be hashed using strong encryption algorithms (e.g. bcrypt).

NFR7 Input Validation:

The system should prevent common attacks such as SQL injection, XSS and CSRF through server-side validation.

NFR8 Secure Payments:

All payment transactions should be performed through SSL-secured HTTPS connections.

Usability Requirements**NFR9 Responsive Design:**

The system interface should be fully responsive and accessible on desktop, tablet and mobile devices.

NFR10 Accessibility:

The system should comply with WCAG 2.1 Level AA standards to support users with disabilities.

Maintainability Requirements**NFR11 Modular Code Structure:**

The system should follow an MVC architecture to facilitate maintenance, debugging and future enhancements.

NFR12 Documentation:

All modules should include inline comments and developer documentation to support easy updates and onboarding.

Data Backup & Recovery Requirements**NFR13 Daily Backups:**

The system should perform automatic database backups daily and store them securely with a minimum 30-day retention.

NFR14 Recovery Time Objective (RTO):

In the event of a system failure, the system should be fully recoverable within 30 minutes.

Privacy & Compliance Requirements

NFR15 Data Privacy Compliance:

The system should comply with the Australian Privacy Principles (APPs) for handling and storing personal data.

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.